A Two-Stage Machine Learning Framework to Support Early Detection of Severe Febrile Illness in Young Children in Resource-Limited Settings

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Abstract

Early and accurate identification of severe febrile illness in young children remains a major challenge in resource-limited settings, where non-specific signs, limited laboratory capacity, and delayed diagnostics hinder timely treatment. This study proposes and evaluates a two-stage machine-learning framework in which a clinical-only "screener" rapidly stratifies risk at first contact, and a biomarker-augmented "specialist" refines classification for indeterminate cases. In a multi-country cohort of 3,379 febrile children aged 1-59 months, the optimal configuration (a Random Forest screener optimised for early severe detection paired with a stacked meta-learner specialist) achieved balanced accuracies of 0.666 (internal validation) and 0.663 (external validation), escalating 80% of severe cases for specialist review on the independent test set. On external validation, mean severe sensitivity increased from 0.232 to 0.444 with the second stage, while maintaining high discrimination for non-severe cases (PR AUC = 0.755). This proof-of-concept demonstrates that staged, data-driven triage may improve early recognition of severe paediatric febrile illness while optimising use of limited diagnostic resources, offering a pragmatic, adaptable approach for frontline care in resource limited settings.

Introduction

Acute febrile illness remains a leading cause of preventable morbidity and mortality in children, with the greatest burden in resource-limited settings where diagnostic capacity is constrained and case mix is complex (Chandna et al., 2021; Valim et al., 2021). In these contexts, frontline clinicians often depend on clinical signs that are variably reproducible and insufficiently sensitive for early identification of high-risk infants (Shafiq et al., 2024). The resulting diagnostic ambiguity is magnified where malaria and bacterial bloodstream infections co-circulate, and where co- or subclinical infections can confound presentation and triage (Krumkamp et al., 2020; Struck et al., 2020).

Despite their ubiquity, current tools have material limitations in accuracy. IMCI-like danger-sign checklists show high specificity but insufficient sensitivity for mortality prediction in young infants (Shafiq et al., 2024). Traditional biomarkers—C-reactive protein (CRP), procalcitonin (PCT), and leukocyte indices—are inconsistently specific early in illness, with performance varying by setting and syndrome (Westwood et al., 2015; Waterfield et al., 2018; Krack et al., 2024). CRP rises slowly in early disease, blunting early rule-in/rule-out utility (Mahendiran et al., 2018), while PCT can outperform white-cell measures for some bacterial–viral differentiations yet is not universally available (Trouillet-Assant et al., 2020). Imaging adds limited reliability in very young infants and may be inaccessible (Florin et al., 2021).

Limitations in speed and infrastructure further constrain care. Blood culture, a de facto reference for invasive bacterial infection, often exhibits suboptimal sensitivity and a reporting delay frequently exceeding 24-48 h, necessitating empiric therapy before confirmation (Heida et al., 2015; Priolo et al., 2023; He et al., 2020). Molecular assays (e.g., PCR) can refine aetiology but require resources rarely available at the point of first contact (Alcoba et al., 2017). Cost and operational complexity remain major barriers to deploying advanced diagnostics at scale in low-income settings (Khadka et al., 2022; Valim et al., 2021).

These constraints motivate pragmatic, staged diagnostics that align information yield with resource cost. This study therefore considers a two-stage machine-learning (ML) framework.

In stage 1, a model trained only on routinely available clinical features (history, vital signs, examination) aims to act as a rapid, low-cost triage to identify children at both very low and the greatest levels of near-term risk, thereby safely reducing laboratory testing and streamlining disposition, whilst aiming to reduce delay in identifying patients who may progress quickly to severe illness. Optimally, in stage 2, for patients flagged as higher (but not immediately critical) risk, a second model aims to integrate a parsimonious subset of laboratory biomarkers (e.g., CRP, PCT, basic haematology) to sharpen discrimination and guide targeted confirmatory testing such as blood culture or PCR. Trained and regularised on representative data, ideally, such models could generalise across epidemiological contexts and be updated as patterns shift (Géron, 2017; Goodfellow et al., 2017; Huyen, 2022). Framed within probabilistic decision theory, sequential testing operationalises explicit trade-offs between diagnostic benefit and resource expenditure (Murphy, 2012).

Here, this framework is applied to a multi-country cohort of febrile children aged 1–59 months recruited from first-contact health facilities across five LMICs in Asia. The performance of multiple ML algorithms across different feature sets was first compared to identify optimal Stage 1 (clinical-only) and Stage 2 (clinical + laboratory) models. Following this, a systematic evaluation of cascade configurations, assessing trade-offs between screening efficiency and severe case sensitivity, was undertaken. Finally, the best-performing configuration was validated on an independent test set. This study addresses three primary research questions: **(i)** what is the relative predictive performance of clinical-only versus integrated clinical–laboratory ML models for classifying SFI outcomes in children in RLS? **(ii)** how might these models be optimally combined into a two-stage cascade to balance patient safety and resource efficiency? and **(iii)** what is the generalisable performance of the optimally identified cascade on unseen data?

Results

Cohort Characteristics

The final analytic cohort comprised 3,379 participants, with stratification by clinical outcome revealing a substantial class imbalance (Table 1). The 'Probable Severe' category was the largest (N = 1,829; 54.1%), while confirmed severe cases were rare (N = 133; 3.9%), necessitating the use of inverse probability weighting in subsequent modelling.

Table 1: Summary Characteristics of Cohort Across Outcomes

Characteristic	Overall (N = 3379)	Non-Severe (N = 638)	Probable Non-Severe (N = 779)	Probable Severe (N = 1829)	Onset > 24h (N = 33)	Onset < 24h (N = 100)	p-value
Age (Months)	16.7 (8.7, 31.1)	22.8 (11.9, 37.5)	17.8 (9.4, 32.2)	15.7 (8.4, 28.6)	3.5 (2.5, 7.5)	6.0 (2.7, 22.8)	<0.001
Site: Rumah Sakit Umum Daerah Wates, Indonesia	186 (5.5%)	69 (10.8%)	30 (3.9%)	87 (4.8%)	0 (0.0%)	0 (0.0%)	<0.001
Site: Angkor Hospital, Cambodia	816 (24.1%)	165 (25.9%)	273 (35.0%)	342 (18.7%)	5 (15.2%)	31 (31.0%)	<0.001
Site: Salavan Provincial Hospital, Laos	205 (6.1%)	84 (13.2%)	58 (7.4%)	63 (3.4%)	0 (0.0%)	0 (0.0%)	<0.001
Site: Savannakhet Provincial Hospital, Laos	77 (2.3%)	29 (4.5%)	15 (1.9%)	33 (1.8%)	0 (0.0%)	0 (0.0%)	<0.001
Site: Dong Nai Childrens Hospital, Viet Nam	938 (27.8%)	136 (21.3%)	121 (15.5%)	649 (35.5%)	3 (9.1%)	29 (29.0%)	<0.001
Site: National Childrens Hospital, Viet Nam	611 (18.1%)	0 (0.0%)	159 (20.4%)	426 (23.3%)	2 (6.1%)	24 (24.0%)	<0.001
Heart Rate (bpm)	146.0 (130.0, 164.0)	136.0 (122.0, 148.0)	143.0 (130.0, 160.5)	150.0 (134.0, 168.0)	167.0 (150.0, 180.0)	170.0 (151.0, 186.0)	<0.001
Respiratory Rate (breaths/min)	38.0 (32.0, 47.0)	35.0 (30.0, 40.0)	37.0 (31.0, 46.0)	40.0 (33.0, 50.0)	60.0 (50.0, 65.0)	50.0 (38.0, 65.0)	<0.001
Oxygen Saturation (%)	98.0 (97.0, 99.0)	99.0 (98.0, 100.0)	98.0 (97.0, 99.0)	98.0 (97.0, 99.0)	98.0 (95.5, 99.5)	97.0 (94.2, 98.0)	<0.001
Capillary Refill Time > 2s	192 (5.7%)	16 (2.5%)	45 (5.8%)	102 (5.6%)	14 (42.4%)	15 (15.0%)	<0.001
SIRS Score	2.0 (1.0, 2.0)	1.0 (1.0, 2.0)	1.0 (1.0, 2.0)	2.0 (1.0, 3.0)	2.0 (1.0, 2.0)	2.0 (1.0, 3.0)	<0.001
Haemoglobin (mg/dL)	11.3 (10.3, 12.2)	11.4 (10.6, 12.2)	11.5 (10.5, 12.3)	11.3 (10.3, 12.2)	10.3 (9.3, 11.9)	10.9 (9.7, 12.1)	<0.001
C-Reactive Protein	14.3 (3.5, 51.4)	8.3 (2.3, 26.6)	13.4 (3.9, 35.8)	17.1 (4.0, 66.1)	9.3 (4.5, 33.7)	35.0 (5.5, 118.2)	<0.001
TNFR1 (pg/ml)	1580.0 (1257.8, 2054.2)	1360.0 (1114.0, 1665.0)	1515.0 (1218.5, 1883.5)	1688.0 (1341.0, 2214.0)	2267.0 (1568.0, 3166.5)	2094.0 (1526.8, 3152.0)	<0.001
Interleukin-6 (pg/ml)	17.9 (7.6, 46.3)	11.3 (5.5, 25.5)	14.8 (6.6, 33.7)	22.0 (9.1, 59.6)	42.1 (8.7, 148.0)	54.3 (12.3, 212.0)	<0.001

*Chi-Square test used for categorical variable significance, Kruskal-Wallis Test employed for continuous variables.
**Categorical characteristics reported as counts with associated proportions, continuous characteristics with their median values, and and associated IQR.
***See supplementary materials for full summary data.

Baseline characteristics varied significantly by outcome. Severe disease occurred in younger participants (median age 3.5–6.0 months vs. 22.8 for non-severe; Kruskal–Wallis, $p < 0.001$). Site of enrolment was also strongly associated with severity (Chi-square, $p < 0.001$), with certain centres contributing disproportionate numbers of severe cases, likely reflecting geographic and health system heterogeneity. Sex was not associated with outcome ($p = 0.39$).

Several clinical signs and laboratory markers indicative of host response and physiological stress were strongly associated with severity ($p < 0.001$ for all variables). Indicators of physiological derangement (e.g., elevated heart rate, prolonged capillary refill) and inflammatory biomarkers (e.g., C-reactive protein, IL-6, TNFR1) displayed progressive increases with severity. For example, median IL-6 rose from 11.3 pg/ml in non-severe cases to 54.3 pg/ml in early-onset severe cases, consistent with escalating systemic inflammation.

Single Stage Model Performance

Table 2: Top Macro Precision-Recall Performing Models Across Feature Sets

Model	Feature Set	Macro PR AUC	Macro ROC AUC	Macro Sensitivity	Macro Specificity	Hyperparameters
Gradient Boosted (XGBoost)	All	0.521	0.852	0.411	0.878	Tree Depth=5; Min. Node Size=13; Learning Rate=0.018
Random Forest	Clinical only	0.504	0.836	0.335	0.859	Predictors per Split=9; Min. Node Size=12
Gradient Boosted (XGBoost)	Laboratory only	0.355	0.736	0.282	0.836	Tree Depth=7; Min. Node Size=16; Learning Rate=0.005
Logistic Regression (Baseline)	Features Present in >= 10% of Literature	0.264	0.661	0.214	0.806	N/A (Not Tuned)

*PR AUC refers to precision-recall area under the curve.
ROC AUC refers to receiver operating characteristic area under the curve.
*Performance metrics shown are the mean values calculated across 25 folds from a 5-fold, 5-repeat stratified cross-validation.

An initial comparative analysis of single-stage models sought to identify the top-performing algorithm for each primary feature set, benchmarked against a baseline logistic regression model (Table 2). A gradient-boosted model (XGBoost) trained on the complete feature set achieved the highest overall performance (macro-averaged PR AUC = 0.521). Notably, a Random Forest classifier restricted to 'Clinical only' features demonstrated highly competitive performance (PR AUC = 0.504), suggesting that a substantial proportion of the predictive signal is contained within readily available clinical variables. All top-performing machine learning models substantially outperformed the baseline (PR AUC = 0.264).

Further, the optimal configuration for predicting each of the five discrete outcome classes individually was identified through consideration of individual-outcome PR AUC metrics (Table 3). While gradient boosted models (XGBoost) using all features were superior for four of the five classes, a marked exception was observed for the 'Onset < 24 hours' severe outcome. For this endpoint, a Random Forest model trained exclusively on clinical data achieved the highest performance (per-class PR AUC = 0.443), indicating that for rapid disease progression, readily available clinical data may contain the most important predictive signals. Substantial variability in class predictability was also exhibited; models showed high discriminatory capacity for 'Non-Severe' (PR AUC = 0.790) and 'Probable Severe' (PR AUC = 0.786) outcomes but performed poorly on the 'Onset > 24 hours' class (maximal PR AUC = 0.228).

Table 3: Highest Outcome Specific Precision-Recall Performing Models

Outcome Class	Best Performing Model	Feature Set	Per-Class PR AUC	Hyperparameters
Non-Severe	Xgboost	All	0.790	Tree Depth=14; Min. Node Size=7; Learning Rate=0.008
Onset greater than 24 hours	Xgboost	All	0.228	Tree Depth=5; Min. Node Size=13; Learning Rate=0.018
Onset within 24 hours	Random Forest	Clinical only	0.443	Predictors per Split=17; Min. Node Size=2
Probable Non-Severe	Xgboost	All	0.458	Tree Depth=11; Min. Node Size=11; Learning Rate=0.007
Probable Severe	Xgboost	All	0.786	Tree Depth=11; Min. Node Size=11; Learning Rate=0.007

*PR AUC refers to precision-recall area under the curve.
*Performance metrics shown are the mean values calculated across 25 folds from a 5-fold, 5-repeat stratified cross-validation.

Following this, multi-objective optimisation was then performed on clinical-only models to identify optimal 'Screener' candidates against conflicting goals of safety and efficiency (Table 4). As such, a Pareto front analysis was performed to explore the explicit trade-off between screening efficiency (identifying non-severe cases) and patient safety (sensitivity to severe cases). At the extremes of this frontier, a Support Vector Machine provided maximal safety (mean severe sensitivity = 0.195), whereas an XGBoost model provided maximal efficiency (Non-Severe PR AUC = 0.764) at the cost of lower sensitivity.

Table 4: Performance Profiles of Pareto-Optimal Screener Models

ML Model Type	Non-Severe PR AUC	Mean 'Severe' Outcome Sensitivity	Sens (Probable Severe)	Sens (Onset > 24h)	Sens (Onset < 24h)	Hyperparameters
Support Vector Machine	0.745	0.195	0.793	0.154	0.237	Cost=24.6; Sigma=0.006
Support Vector Machine	0.745	0.192	0.786	0.169	0.215	Cost=11.7; Sigma=0.01
Gradient Boosted (XGBoost)	0.747	0.141	0.812	0.062	0.220	Tree Depth=14; Min. Node Size=6; Learning Rate=0.028
Gradient Boosted (XGBoost)	0.763	0.129	0.844	0.054	0.205	Tree Depth=3; Min. Node Size=5; Learning Rate=0.016
Gradient Boosted (XGBoost)	0.764	0.090	0.844	0.031	0.149	Tree Depth=5; Min. Node Size=11; Learning Rate=0.012

*PR AUC refers to precision-recall area under the curve.

**Performance metrics shown are the mean values calculated across 25 folds from a 5-fold, 5-repeat stratified cross-validation.

These findings informed the assembly of six Stage 1 screeners: the five Pareto front models spanning the safety-efficiency spectrum, plus the clinical-only Random Forest best for 'Onset < 24 hours'. For Stage 2 specialists, the overall best all-feature XGBoost and a 'Meta-Learner Panel' combining per-class best models (Table 3) were selected. All cascades were tested both with and without a clinical safeguard policy that retained any Stage 1 severe prediction without handoff.

Two-Stage Cascade Performance

As such, comparative analysis of the cascade configurations suggests that employing the Meta-Learner as the Stage 2 specialist yields superior performance, particularly in achieving the highest balanced accuracy and mean severe sensitivity (Figure 1). The application of the Safeguard policy also consistently improves performance across key metrics. However, these evaluations highlighted clear performance trade-offs, where models optimised for one metric often exhibited deficiencies in others, such as severe case sensitivity, necessitating a structural analysis of performance.

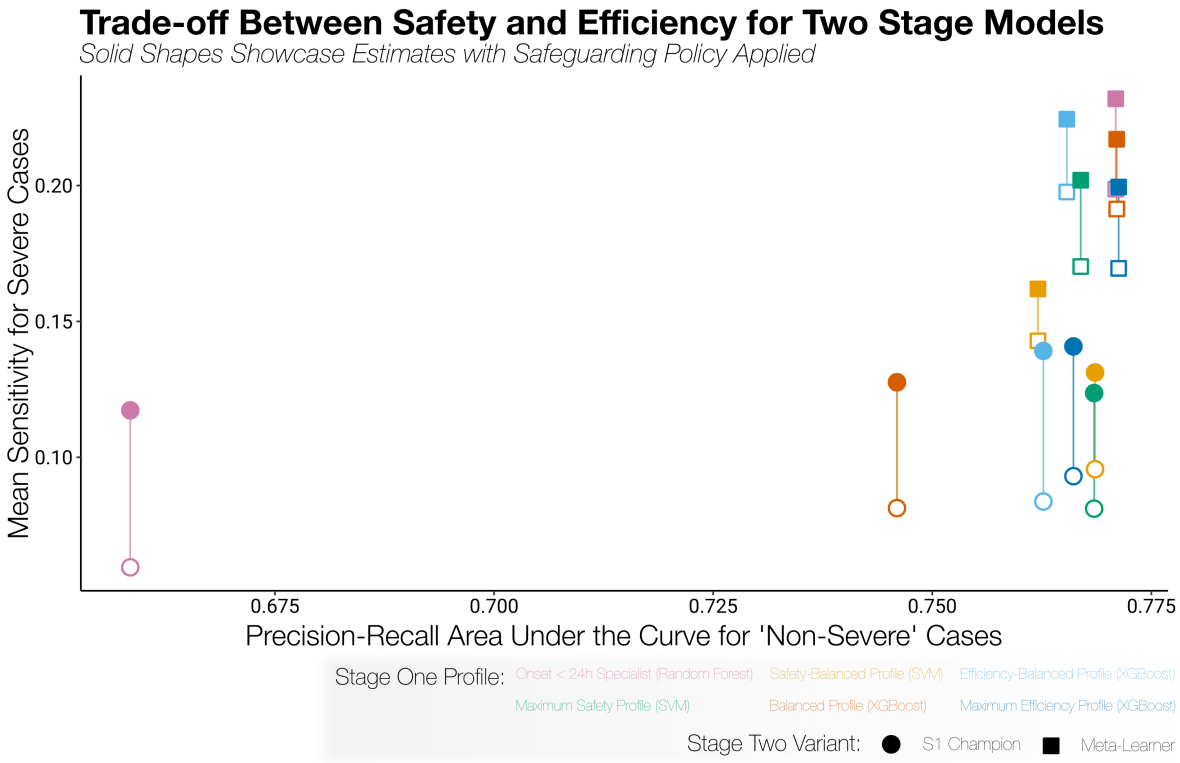

Figure 1: Safety Versus Efficiency Across Cascades

To deconstruct these interdependencies, a principal component analysis (PCA) was conducted on the performance metrics across cascade configurations (Table 5). The first three principal components (PCs) were found to explain approximately 94% of the total variance in model performance. The leading component (PC1), suggests an axis of general model performance, where lower scores correspond to higher values across nearly all metrics, but is confounded by an association with higher handoff rates. PC2 addresses this conflict, representing a potential axis of cascade efficiency, where a higher score simultaneously corresponds to higher sensitivity and lower handoff rates. Finally, PC3 present a potential clinically relevant trade-off between overall efficiency and severe case confidence, contrasting the overall handoff rate with the more critical severe case handoff rate.

Based on this, the definitive cascade model selected was the configuration pairing the 'Random Forest - Onset < 24h Specialist' screener with the Meta-Learner specialist under the Safeguard policy. This model was selected not only for achieving the highest balanced accuracy (0.666), highest mean severe sensitivity (0.232), and lowest severe case handoff rate

(76.4%) among all evaluated strategies, but also for its superior PCA profile. Its strong negative PC1 score aligned with a higher overall quality, while a positive PC2 score demonstrated an optimal balance of sensitivity and efficiency. Furthermore, a slightly positive PC3 score reflects its superior performance in confidently classifying severe cases with minimal handoff, making it the most robust and clinically aligned configuration (*see supplementary materials, Annex B, for further information*).

Table 5: Loadings on First Three Principal Components of Final Results

	PC1	PC2	PC3
% Severe Case Handoff Rate	-0.142	-0.492	-0.843
% Handoff Rate	-0.175	-0.573	0.299
Avg. Severe Sensitivity	-0.288	0.418	-0.270
S2 Net Corrections	-0.330	-0.370	0.263
Macro PR AUC	-0.381	0.127	0.144
PR AUC (Non-Severe)	-0.382	-0.090	0.124
Macro Sensitivity	-0.384	0.233	-0.103
Balanced Accuracy	-0.393	0.198	-0.079
Macro Specificity	-0.406	-0.022	0.066

Final Cascade Performance on Held-Out Test Data

To provide an unbiased estimate of its generalisable performance, the definitive two-stage cascade model was deployed on an unseen test dataset (N = 677). The final configuration consisted of a Random Forest screener optimised for early severe case detection, a stacked meta-learner specialist, and a clinical safeguard policy (Table 6).

The final model achieved a balanced accuracy of 0.663 on the test set. It demonstrated high efficiency in identifying non-severe cases, yielding a PR AUC of 0.755 for the 'Non-Severe' class. For the primary safety objective, the model achieved a mean sensitivity of

0.444 across the two confirmed severe outcome classes. The overall macro-averaged PR AUC was 0.245,reflecting the difficulty of predicting the rarest outcome classes with high precision and recall simultaneously.

Analysis of the cascade mechanics reveals a high reliance on the specialist tier, with a total handoff rate of 91.7%. Critically, 80.0% of all confirmed severe cases in the test set were correctly escalated for specialist review; the effective severe case handoff rate was also 80.0% (defined as the proportion of cases escalated despite the screener's initial prediction being for a non-severe outcome). The equivalence of these two rates demonstrates that every severe case escalated to Stage 2 was one that the Stage 1 screener had provisionally misclassified as non-severe. This finding underscores a tradeoff of the handoff rule in its primary safety-oriented role: identifying and escalating the cascades own most critical potential errors.

After applying the safeguard policy, the Stage 2 meta-learner resulted in a net change of -1 correct classifications. An analysis of the system without this policy revealed that the meta-learner would have made a net of +1 correct classifications. This inversion demonstrates a potential downside of the clinical safeguard rule, which, in its function to mitigate specific high-risk misclassification errors, overrode at least one correct re-classification made by the specialist model to ensure maximal patient safety.

Table 6: Summary of Final Cascade Model Performance on Test Data

Metric	Value
Stage One Model	Random Forest - Onset < 24h Specialist
Stage Two Model	Meta-Leaner
Policy	Safeguard
Balanced Accuracy	0.663
Avg. Severe Sensitivity	0.444
PR AUC (Non-Severe)	0.755
Macro PR AUC	0.245
Macro Sensitivity	0.451
Macro Specificity	0.876
% Handoff Rate	91.7
% Effective Handoff Rate	91.4
% Severe Case Handoff Rate	80
% Effective Severe Case Handoff Rate	80
S2 Net Corrections (with safeguard)	-1
S2 Net Corrections (if safeguard ignored)	1

Methods

Spot Sepsis Data

This analysis used data from the Spot Sepsis cohort, a multi-country, prospective study conducted across seven hospitals in Bangladesh, Cambodia, Indonesia, Laos, and Viet Nam between March 2020 and November 2022, the protocol for which is described in detail elsewhere (Chandna et al., 2021, 2025).

The study enrolled 3,423 children aged 1–59 months presenting with a community-acquired febrile illness of $\leq$ 14 days' duration. Key exclusion criteria included prior hospital admission or receipt of significant parenteral treatment for the current episode, recent routine immunisations, trauma, or specific chronic comorbidities such as known immunosuppression or active cardiorespiratory disease (Chandna et al., 2021, 2025).

At enrolment, trained study personnel collected structured clinical data, vital signs, and biological samples for biomarker analysis and molecular pathogen detection. Participants were followed up on days 2 and 28 post-enrolment to ascertain clinical outcomes, with standardised definitions applied across all sites.

Literature Informed Baseline Model

A systematic literature review was undertaken to establish a benchmark model for subsequent machine learning (ML) comparisons. Peer-reviewed, English-language articles published between January 2005 and April 2025 were identified through Scopus, PubMed, Web of Science, and Google Scholar, yielding 155 studies applying statistical or ML modelling to severe febrile illness (SFI) in children $\leq$ 5 years (Figure 2).

From these, predictors were extracted and grouped into three frequency-based tiers. Tier 1 variables, appearing in $\geq$ 10% of studies, were selected for baseline inclusion owing to both prevalence and availability in the Spot Sepsis dataset.

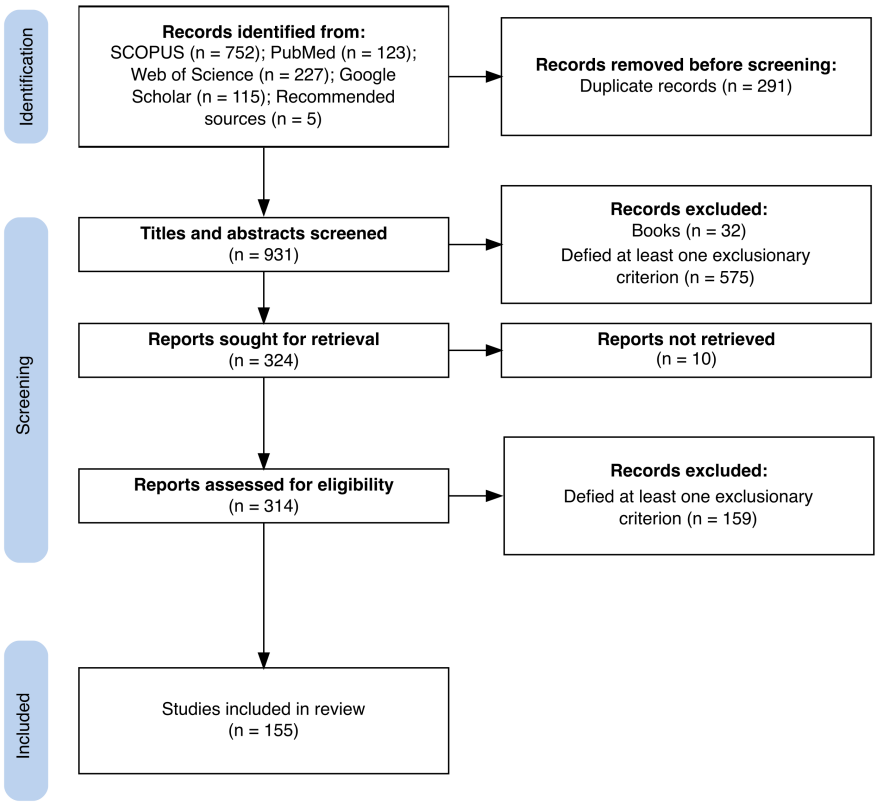

Figure 2: PRISMA Screening

A Directed Acyclic Graph (DAG) was developed to formalise causal relationships between SFI, primary biomarkers (C-reactive protein, procalcitonin), and potential confounders. This identified 'age' and 'underlying infection' as the minimal adjustment set required for unbiased estimation of the biomarker–SFI association, yielding the simplified logistic regression model in (1).

$$\text{Severe Febrile Illness} \sim \text{C-reactive Protein} + \text{Procalcitonin} + \text{Age} + \text{Underlying Infection} \quad (1)$$

The baseline model was implemented in R using the tidymodels framework. Preprocessing comprised reverse-probability case weighting, median/mode imputation, one-hot encoding, and normalisation. Model performance was evaluated via 5-fold cross-validation, repeated five times, with metrics (e.g., ROC AUC, PR AUC) averaged across resamples. This baseline provided a benchmark against which more complex ML algorithms were assessed. Detailed search strategies, the DAG, and full predictor lists are provided in the Supplementary Information.

While most reviewed studies employed descriptive statistics or logistic regression, a smaller subset used ML approaches (Random Forests, 5.8%; penalised regression, 3.9%; CART, 3.2%; SVM, 2.6%; gradient-boosted trees, 0.6%). These informed selection of five benchmark algorithms: a 1,000-tree Random Forest, Elastic Net, CART, Support Vector Machine, and XGBoost.

Outcome Definition and Data Preprocessing

The outcome was defined as a five-level ordinal variable. The original 'Severe disease' category was sub-stratified by time-to-event: 'Onset $\leq$ 24 hours' and 'Onset $>$ 24 hours' post-enrolment. This, combined with the existing 'Probable Severe', 'Probable Non-Severe', and 'Non-Severe' categories, aimed to capture temporal heterogeneity in disease progression. Class imbalance was exacerbated by this stratification, and was attempted to be addressed by computing inverse probability case weights, applied throughout model training.

A multi-stage preprocessing pipeline ensured data integrity and compatibility across algorithms. Steps included:

1. Assignment of case weights.
2. Creation of binary missingness indicators for each predictor.
3. Median imputation for numeric and mode imputation for categorical variables.
4. One-hot encoding for all models except single decision trees.
5. Normalisation for linear and kernel-based models.
6. Discretisation of continuous variables into deciles for the single decision tree.
7. Training and Evaluation of Benchmark Algorithms

Training and Evaluation of Benchmark Algorithms

Hyperparameter tuning for each ML model used a standardised grid or random search of 64 parameter combinations, evaluated against the same 5×5 stratified cross-validation folds to ensure comparability (see Supplementary Information, Annex C).

To isolate the contribution of different data types, models were trained and evaluated on three feature sets:

1. **Clinical Only**: predictors from patient history and examination (simulating resource-limited assessment).
2. **Laboratory Only**: laboratory-derived predictors (e.g., CRP, procalcitonin, lactate).
3. **All Predictors**: combined clinical and laboratory features.

This tiered strategy allowed direct assessment of the incremental value of laboratory data over clinical information.

Designing the Two-Stage Cascade Framework

Following single-stage analysis, a cross-validated, performance-driven process was used to select candidate models for a two-stage cascade framework, designed to pair computationally efficient Stage 1 'Screener' models with higher-capacity Stage 2 'Specialist' models.

Candidates for the Stage 2 models were selected from models trained on the complete feature set (clinical and laboratory data) to maximise predictive power. Two distinct specialist strategies were established based on performance: (1) The overall top-performing model across all analyses, ranked by macro-averaged area under the precision-recall curve (PR AUC), was selected. This was identified as an XGBoost classifier; (2) to leverage the unique predictive strengths of different algorithms, a panel was assembled comprising the five models that each achieved the highest per-class PR AUC for one of the five distinct outcome classes. The predictions from this panel were designated for use as input features to a stacked meta-learner.

For selection of the stage 1 screener, models were trained on clinical-only features for rapid deployment feasibility. Multi-objective optimisation was used to identify models balancing patient safety (average sensitivity across the two most severe outcomes) and screening efficiency (non-severe PR AUC). Pareto front analysis yielded five optimal models spanning this trade-off spectrum. A sixth screener was added: the top-performing clinical-only specialist for 'Probable Non-Severe'.

Evaluation of Cascade Configurations

Twelve cascade configurations (six screeners $\times$ two specialists) were evaluated using the same 25 cross-validation folds from earlier analyses. Within each fold, data were split into training (80%) and calibration (20%) sets. Stage 1 screeners were trained on the training partition, and probabilistic outputs were calibrated via isotonic regression on the calibration set.

A dynamic handoff rule was optimised on the calibration set using a two-dimensional grid search over confidence thresholds for two anchor outcomes: 'Onset $\leq$ 24 hours' and 'Non-Severe'. Predictions exceeding these thresholds were resolved by the screener; others were escalated to Stage 2. This process aimed to maximise predictive accuracy on the high-confidence subset while managing the trade-off between coverage and precision.

The Champion specialist was trained on all features from escalated cases only. The Meta-Learner specialist involved: **(i)** training the five-model panel on the training partition; **

(ii) ** generating calibration-set predictions; ** (iii) ** applying PCA to retain components explaining $\geq 90\%$ of variance; and ** (iv) ** training a multinomial logistic regression meta-learner on the resulting components.

A clinical safeguard policy was also evaluated, in which any Stage 1 'severe' prediction was retained without specialist reclassification (2).

$$\text{If } \text{Pred}_1 \in \text{Severe and } \text{Pred}_2 \notin \text{Severe}, \text{ then } \text{Pred}_{\text{Final}} = \text{Pred}_1 \tag{2}$$

Final Model Construction and Deployment

After comparative cross-validation, the final cascade was trained on a fixed 80% training / 20% calibration split. The Stage 1 screener (Random Forest) was trained on 34 clinical predictors. The calibration set was used to fit isotonic regression for probability calibration and to finalise the static handoff rule via the grid search process described above.

The Stage 2 specialist (four-model panel) was trained on all 52 clinical and laboratory variables. Predictions were transformed using PCA before input to the final multinomial logistic regression meta-learner.

For unbiased evaluation, the fixed cascade was deployed on the previously unseen test dataset. Each case was processed through Stage 1, triaged via the static handoff rule, escalated where applicable to Stage 2, and subjected to the safeguard policy. The safeguarded output formed the basis for final performance reporting.

Discussion

This multi-country study evaluated a two-stage machine learning (ML) cascade for classifying severe febrile illness (SFI) in 3,379 children aged 1–59 months presenting to first-contact health facilities across five Asian LMICs. The framework, combining a clinical-only Stage 1 "screener" with a biomarker-augmented Stage 2 "specialist," achieved a balanced accuracy of 0.666 in internal validation and 0.663 in external validation, escalating 80% of severe cases for specialist review on the independent test set. Stage 1 demonstrated competitive discrimination using only readily available clinical features (best PR AUC = 0.504), while Stage 2, optimally represented by a meta-learner combining per-class best models, improved classification of indeterminate-risk cases. On external validation, mean severe sensitivity rose from 0.232 to 0.444, with high non-severe discrimination and reduced laboratory testing demand, underscoring the framework's potential to balance safety with resource efficiency.

These findings directly address a recognised public health challenge: the difficulty of identifying children at risk of severe outcomes in LMIC primary care and community settings (Chandna et al., 2021; Ogero et al., 2020). The cascade extends beyond the clinical prediction models (CPMs) and scoring systems predominant in the literature, offering a flexible, data-driven approach that can be adapted to local constraints.

Stage 1 results confirm both the value and limitations of clinical features alone. While they effectively identified substantial low-risk groups, overall discrimination remained imperfect. This is consistent with previous observations that such features may flag potential severity but leave residual diagnostic uncertainty (Van den Bruel et al., 2010). Stage 2 integration of laboratory biomarkers builds on evidence that immune and endothelial activation markers can risk-stratify febrile illness regardless of aetiology, particularly for patients whose severity is not clinically apparent at presentation (Van den Bruel et al., 2011); the superior performance of the presented Stage 2 meta-learner for refining intermediate-risk cases supports their role in improving triage accuracy.

Importantly, the cascade's external test-set validation addresses a methodological gap, wherein Ogero et al. (2020) emphasised the paucity of prognostic models for LMIC settings

that undergo such validation on data covering multiple geographies. It is important that these approaches are escalated and validated further on even more diverse data. The inclusion of a clinical safeguard policy operationalises recommendations that prediction tools function as an adjunct to clinical assessment (Chandna et al., 2021), ensuring high-risk predictions are preserved even when algorithmic reclassification suggests otherwise. Although this occasionally overrode correct specialist reclassifications, its role in preventing critical errors reflects an appropriate prioritisation of patient safety in high-stakes environments.

This study's methodological rigour is a key strength. Development followed a systematic review foundation to align with identified evidence gaps, employed robust cross-validation, used a deliberately heterogeneous multi-country cohort, and compared multiple algorithms and feature sets. Principal component analysis enabled multidimensional assessment of trade-offs between safety and efficiency; critical for decision-support tools in constrained health systems.

Nonetheless, several limitations must be considered. The retrospective design limits control over unmeasured confounding, and the absence of follow-up beyond the study period precludes verification of longer-term outcomes. Severe class imbalance and the small number of confirmed severe cases (highly geographically clustered) introduce the high possibility of site-specific effects influencing performance. While inverse probability weighting and site heterogeneity were addressed in training, direct extrapolation to markedly different contexts should be approached with caution. The dataset's restriction to five Asian LMICs also limits direct applicability to other regions. Finally, the modest sample size underscores the need for pooled paediatric datasets to strengthen stability and generalisability.

Taken together, these results provide a strong proof-of-concept for a two-stage cascade framework in paediatric SFI triage, but also highlight avenues for refinement and scaling. A priority next step is the aggregation of complementary datasets to create a substantially larger, cohesive global paediatric sepsis dataset. This would enable more robust model training, improve handling of rare severe phenotypes, and support region-specific calibration strategies.

Looking forward, emerging advances in large language models (LLMs) and multimodal machine learning could enhance both the predictive backend and the user-facing interface. Such

approaches could integrate diverse data streams, including structured clinical variables, laboratory results, and unstructured narratives, while supporting adaptable, multilingual deployment at the point of care (Clusmann et al., 2023; Raza et al., 2025). Recent releases of powerful open-source LLMs further increases the feasibility of developing tailored, regulated tools for high-stakes applications such as SFI triage in resource-limited settings (OpenAI, 2025). Embedding these advances within a continuously updated decision-support ecosystem could help maintain safety and efficiency while allowing models to adapt dynamically to shifting epidemiological and operational realities.

Data Availability

The Spot Sepsis study is registered at ClinicalTrials.gov under accession code NCT04285021. De-identified individual participant data, along with the study protocol, informed consent materials, standard operating procedures, and data management plan, are available upon reasonable request to the Médecins Sans Frontières and Mahidol-Oxford Tropical Medicine Research Unit Data Access Committees (data.sharing@london.msf.org, datasharing@tropmedres.ac). Requests for biobanked samples should be directed to the corresponding author, who will coordinate with the Spot Sepsis Sample Use Committee.

Code Availability

All code used for the analyses in this study is available at https://github.com/ffr0517/spot-sepsis-cascade. No patient-identifiable data are included.

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Author Contributions

- **Conceptualization** – L.C.R., Y.L., A.C.
- **Methodology** – L.C.R., S.S., A.S.K., Y.L., A.C.
- **Software** – L.C.R.
- **Formal analysis** – L.C.R.
- **Investigation** – L.C.R.
- **Resources** – A.C. (Spot Sepsis dataset access), Y.L., S.S.
- **Data curation** – L.C.R., A.C. (primary Spot Sepsis dataset)
- **Writing – original draft** – L.C.R.
- **Writing – review & editing** – L.C.R., S.S., Y.L., A.S.K., A.C.
- **Visualization** – L.C.R.
- **Supervision** – S.S., Y.L.
- **Project administration** – L.C.R., S.S., Y.L.
- **Funding acquisition** – Y.L. (Spot Sepsis study), S.B. (Spot Sepsis study)

Note:

This dissertation uses secondary analysis of data from the Spot Sepsis study. Full author contributions for the primary study are detailed in the original publication (ClinicalTrials.gov accession code NCT04285021) and include, among others, A.C., C.K., R.A.A., P.T., M.M., E.A.A., E.A., R.P.-S., M.R.-G., Y.L., and S.B.

Competing Interests

The author(s) declare that no competing interests exist.

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